Abstract:

Decoherence-free subspaces and their generalization, decoherence-free subsystems (both refered to as DFSs here), are an alternative to quantum error correction (QEC) for dealing with the problem of performing quantum computation in the presence of a noise inducing environment. Proper encoding of quantum information in a DFS sheilds the information from the effects of decoherence. Here we discuss specific forms of decoherence in which the system couples collectively to the bath. We first induce an analogue of the QEC stabilizers for DFSs and show how this can be used to identify properties of the of DFS. We then show that this allows for fault-tolerant quantum computation on the collective DFSs. The fault tolerant computation on the collective DFS can be performed with one and two-qubit interactions which *always preserve the integrity of the quantum information*. This natural fault tolerance is possible due to a duality between DFSs and QEC. The results of this study also have immediate consequences for computation using the exchange interaction.

Subspaces and Subsystems:

E. Knill, R. Laflamme and L. Viola,
Phys. Rev. Lett. 84, 2525 (2000)



Interaction causing DECOHERENCE

System evolution Bath evolution

$$H = H_S \otimes I_B + I_S \otimes H_B + \sum_\alpha S_\alpha \otimes B_\alpha$$

← Expansion into system and bath operators

The S_α form an algebra (add, multiply the elements) which is reducible:

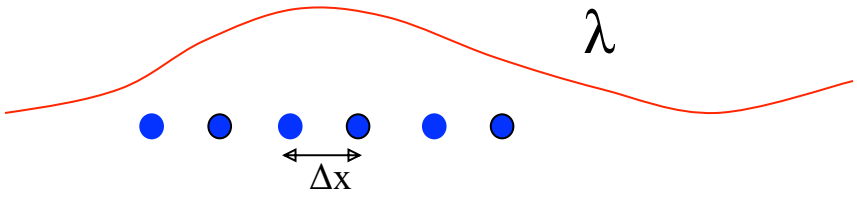
$$\mathcal{A} = \bigoplus_{J=1}^{r} \left(I_{n_J} \otimes M_{d_J} \right)$$

← linear operators on a d_J dimensional space

I_d is the $d \times d$ identity operator n_J is the degeneracy of the Jth irrep

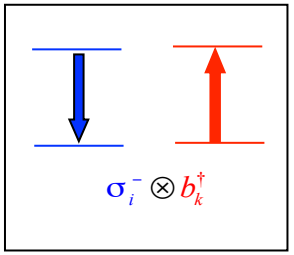
Hilbert space over the degenerate identity component is immune to the effects of decoherence! The quantum information then inhabits a decoherence-free subsystem (subspace if $d_J = 1$).

Collective Decoherence:

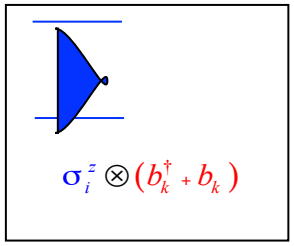


System Bath Interaction (spin-boson model): - Dissipative

$$H_{S-B} = \sum_{i=1}^{N} \sum_{\mathbf{k}} \underbrace{g^{+}_{i,\mathbf{k}} \sigma^{+}_{i} b_{\mathbf{k}} + g^{-}_{i,\mathbf{k}} \sigma^{-}_{i} b^{\dagger}_{\mathbf{k}}}_{\text{dissipation}} + \underbrace{g^{z}_{i,\mathbf{k}} \sigma^{z}_{i} \left(b_{\mathbf{k}} + b^{\dagger}_{\mathbf{k}} \right)}_{\text{dephasing}}$$



- Conservative (dephasing)



$\{\sigma_i^+, \sigma_i^-, \sigma_i^z\}$ satisfy local (i) angular momentum *su(2)* algebra

Collective Interaction Approximation:

bath does not distinguish system particles

$$\boxed{g_{i,\mathbf{k}} \equiv g_{\mathbf{k}}}$$ ← $\lambda \ldots \Delta x$

particle permutation invariance

Define **collective** "angular momentum" operators:

$$S^{\alpha} = \sum_{i=1}^{N} \sigma_i^{\alpha} \qquad \alpha = +, -, z$$

satisfy **global** angular momentum *su(2)* algebra

- strong collective decoherence: all S^{α} present
- weak collective decoherence: only dephasing S^{z}

Rewrite Hamiltonian:

$$H_{S-B} = S^{+} \otimes \left(\sum_{\mathbf{k}} g^{+}_{\mathbf{k}} b_{\mathbf{k}} \right) + S^{-} \otimes \left(\sum_{\mathbf{k}} g^{-}_{\mathbf{k}} b^{\dagger}_{\mathbf{k}} \right) + S^{z} \otimes \left(\sum_{\mathbf{k}} g^{z}_{\mathbf{k}} \left(b_{\mathbf{k}} + b^{\dagger}_{\mathbf{k}} \right) \right)$$

Strong Collective DFS Examples:

3 qubit strong collective decoherence, decoherence-free subsystem:

S^x over the degenerate portion $= \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}$

$\rightarrow I_2 \otimes s_x^2$

$$\left\{\begin{aligned} \left|J=\tfrac{1}{2},m_J=\tfrac{1}{2}\right\rangle &= \tfrac{1}{\sqrt{2}}\left(|01\rangle-|10\rangle\right)|0\rangle \\ \left|J=\tfrac{1}{2},m_J=-\tfrac{1}{2}\right\rangle &= \tfrac{1}{\sqrt{2}}\left(|01\rangle-|10\rangle\right)|1\rangle \end{aligned}\right.$$

$$\left\{\begin{aligned} \left|J=\tfrac{1}{2},m_J=\tfrac{1}{2}\right\rangle &= \sqrt{\tfrac{2}{3}}|00\rangle|1\rangle-\sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle+|10\rangle\right)\right\}|0\rangle \\ \left|J=\tfrac{1}{2},m_J=-\tfrac{1}{2}\right\rangle &= \sqrt{\tfrac{2}{3}}|11\rangle|0\rangle-\sqrt{\tfrac{1}{3}}\left\{\tfrac{1}{\sqrt{2}}\left(|01\rangle+|10\rangle\right)\right\}|1\rangle \end{aligned}\right.$$

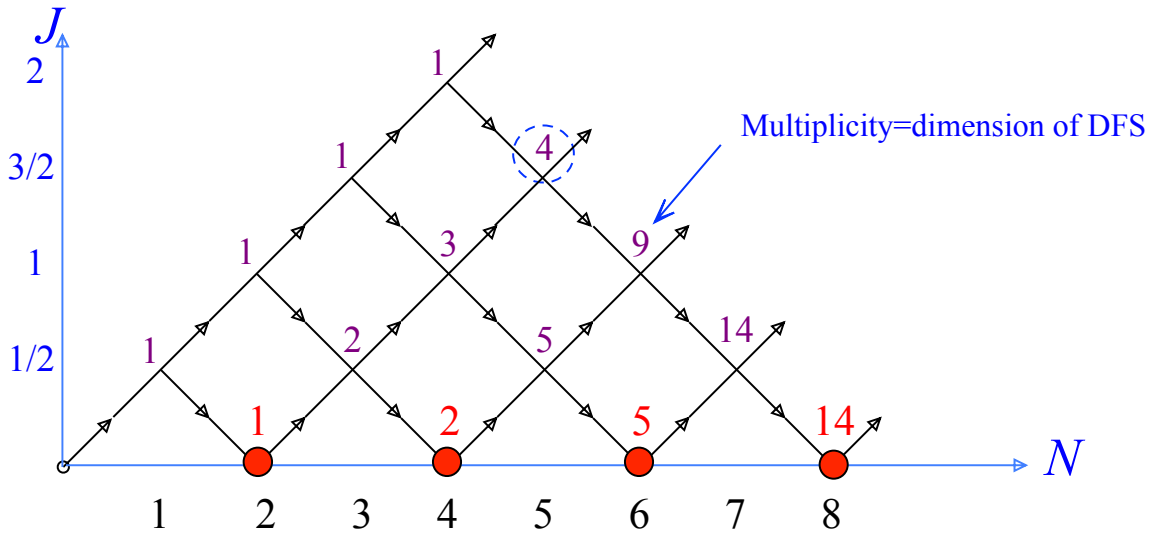
Information encoded over the degenerate component will not decohere!

4 qubit collective decoherence, decoherence-free subspace:

$$\left|J=0,m_J=0,n=0\right\rangle = \tfrac{1}{2}\left(|01\rangle-|10\rangle\right)\otimes\left(|01\rangle-|10\rangle\right)$$

$$\left|J=0,m_J=0,n=1\right\rangle = \tfrac{1}{\sqrt{3}}\left[|0011\rangle+|1100\rangle-\left(|01\rangle+|10\rangle\right)\otimes\left(|01\rangle+|10\rangle\right)\right]$$

Information encoded over the subspace will not decohere

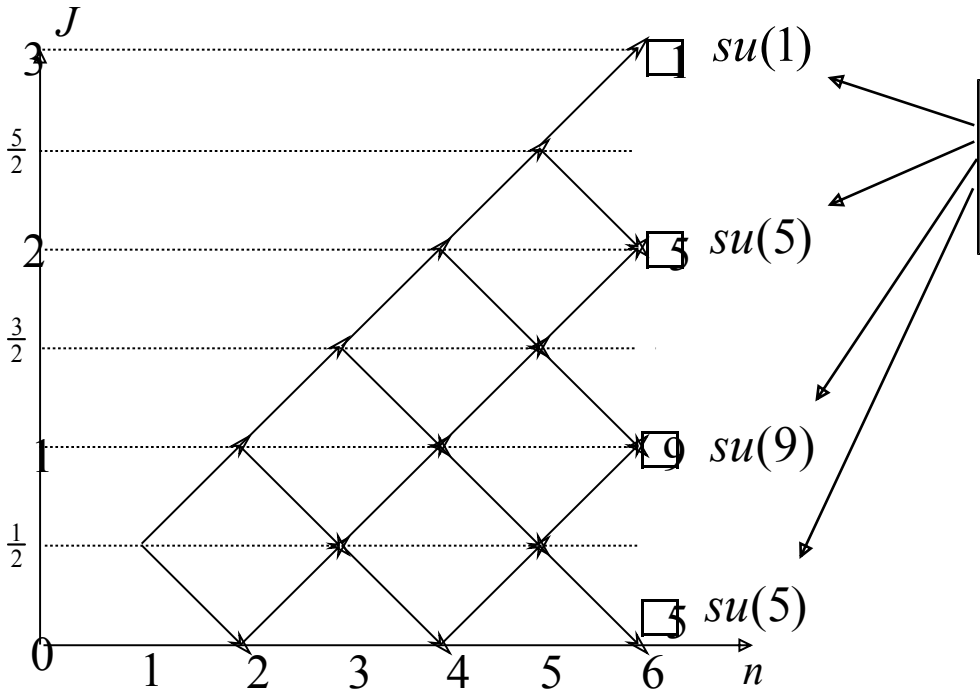


Weak Collective DFSs

Gates for computation:

Here we first state what fault-tolerance means for DFS and QECC:
(never leaves the DFS versus in QECC at beginning and end)
Also we impose the locality requirement on the operations.
Then we use the commutant to get Exchange.

Results:



Exchange can be used to perform su(d)'s on each of the DFSs.

Each su(n) can be inacted when the others turned off (independent action on each J irrep)

su(d) means we have a d dimensional Hilbert space (encoded) and we can perform all unitary manipulations on this d dimensional Hilbert space!

Exchange produces fault tolerant quantum computation on collective decoherence-free subsystems!!!

Asymptotic Universality:

$$\text{encoding efficiency} \equiv r = \frac{\log_2(\text{dimension of DFS})}{\text{n}}$$

As $n \to \infty$, $r(n) \to 1 - c\dfrac{\log_2 n}{n}$. The encoding efficiency is nearly unity for large n!

From a computational perspective working on the encoded space is equivalent to working on a full quantum computer: asymptotically universal.

The Exchange Hamiltonian is ASYMPTOTICALLY UNIVERSAL

Concatentation

Exchange interactions:

The Heisenberg interaction of spin-spin coupling

$H_{Heis} = J(t)s^{(i)}\cdot s^{(j)}$ where $s^{(i)} = \left(s_x^{(i)}, s_y^{(i)}, s_z^{(i)}\right)$ are the single spin operators on qubit i

- Quantum dot quantum computer
- Kane solid state quantum computer

The Heisenberg interaction is exactly the exchange coupling:

$$H_{Heis} = \frac{1}{4}J(t)\left(E_{ij} + I\right)$$

Fast, robust two-qubit operation. Tunable by local magnetic fields with an on-off ratio of several orders of magnitude.

Encoding into lowest J DFS is easily done by cooling the system with a exchanges between alternating neighboring dots.

There is no need to supplement the proposals by single qubit gates

QECC versus DFS:

	Quantum Error Correcting Code for independent decoherence	Decoherence-Free Subsystems for collective decoherence
errors	local	local
error coupling to bath	local	global
mode	active	passive
fault tolerance	must preserve code between gate operations	must not leave the code during gate operations
fault tolerance results	must leave codespace during one and two body interactions	natural fault tolerance provided by the two body exchange interaction

Summary and Conlusions:

Mention recent work with DiVincenzo. Mention that similar method can be used to analyze power of other given interactions. Introduce this whole philosophy.
Mention Euler angles and 19 gate solution.

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References:

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Julia Kempe, Dave Bacon, Daniel Lidar, K. B. Whaley, "Theory of Decoherence-Free Fault-Tolerant Universal Quantum Computation." quant-ph/0004064. (2000)

David DiVincenzo, Guido Burkard, Julia Kempe, Dave Bacon, work in progress (2000)